

NIM

Simple Rules

Start with a number of stones (say 22). Two players take alternating turns removing 1,2,3,or 4 stones on each turn. The player who removes the last stone loses. Who has the winning strategy?

Winning Strategy

- A set of instructions that one of the players can follow in order to ensure his or her victory no matter what the other player does.
- In some of our games, we will discover a winning strategy for the player who goes first, and in others, the player who goes second.

Strategy for Simple Rules

- Player 1 has a winning strategy
- We want to leave our opponent with 1 stone. If there are 2-5 stones left, the player whose turn it is wins.
- If there are 6 stones left, the player whose turn it is loses. No matter what he plays, the other player can play (5 minus that number) and leave him with 1 stone.
- If there are 7-10 stones left, the player whose turn it is wins as he can just leave the opponent with 6 stones.
- If there are 11 stones left, the player whose turn it is loses. No matter what he plays, the other player can play (5 minus that number) and leave him with 6 stones.
- Working backwards like this, we see that players should aim to leave their opponents with a number that has remainder 1 when divided by 5 (this would be 1, 6, 11, 16, 21).
- Player 1's winning strategy is to start with taking 1 stone to leave 21 stones for player 2. Whatever number x Player 2 then removes, Player 1 removes $5-x$.

Variations

- Who wins and what is the strategy if you can only take either 1 or 2 stones? (start with 22 stones)
 - Player 2 wins. Always leave a number that's 1 more than a multiple of 3
- Who wins and what is the strategy if the winner is the one who takes the last stone? (start with 22 stones and you are allowed to take up to 4 stones each turn)
 - Player 1 wins. Always leave a number that's a multiple of 5